Assignment 4 Forms and Validation

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HTML + Bootstrap | Individual assignment | 18 October 2026

GitHub repository URL: PENDING - add the real repository link before submission. This report is not submission-ready without it.

1. Book a train ticket

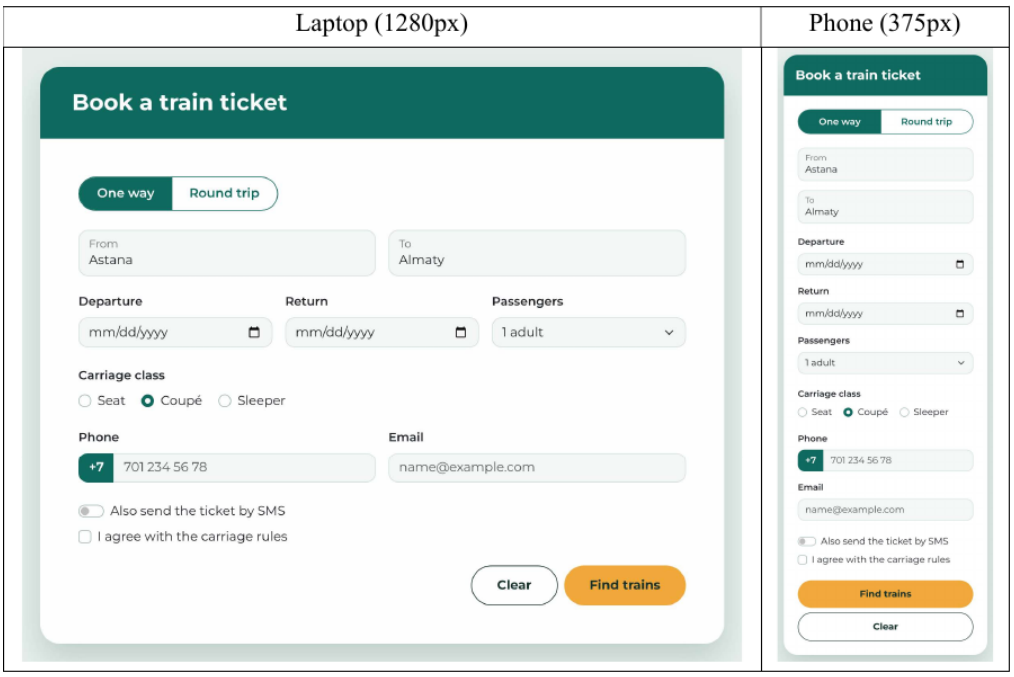
I used Bootstrap rows and responsive columns to arrange the booking fields. The From and To controls have a small label above the input inside one rounded area. Journey type and carriage class use radio inputs, SMS uses a switch, and agreement uses a checkbox. The supplied colors and locally stored Montserrat font recreate the reference style.

The required controls are From, To, Departure, Carriage class, Phone, Email and agreement. Passengers is a required select with options for 1-6 adults. Return is optional. Every control has a connected label. Native controls and visible focus indicators support keyboard use.

The HTML places Clear before Find trains. flex-column-reverse shows Find trains first on small screens; flex-md-row restores the desktop layout. No custom fixed widths or absolute positioning are used.

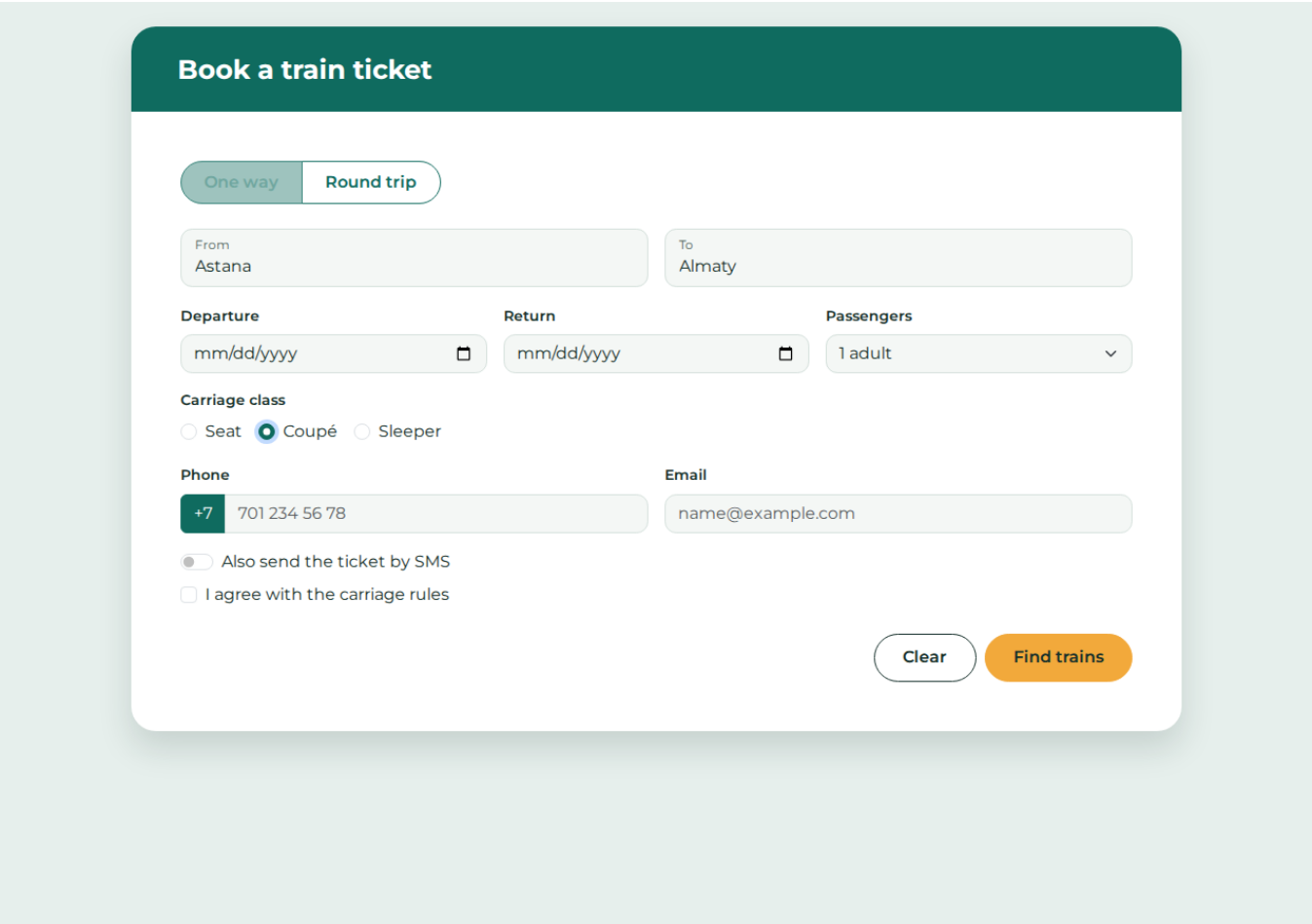
Clear uses type="reset". Fields have empty defaults and no checked radio defaults, so reset really clears them. For the screenshots I entered Astana and Almaty, selected One way, Coupé and 1 adult to match the example. These are demonstration values, not HTML defaults.

Assignment reference

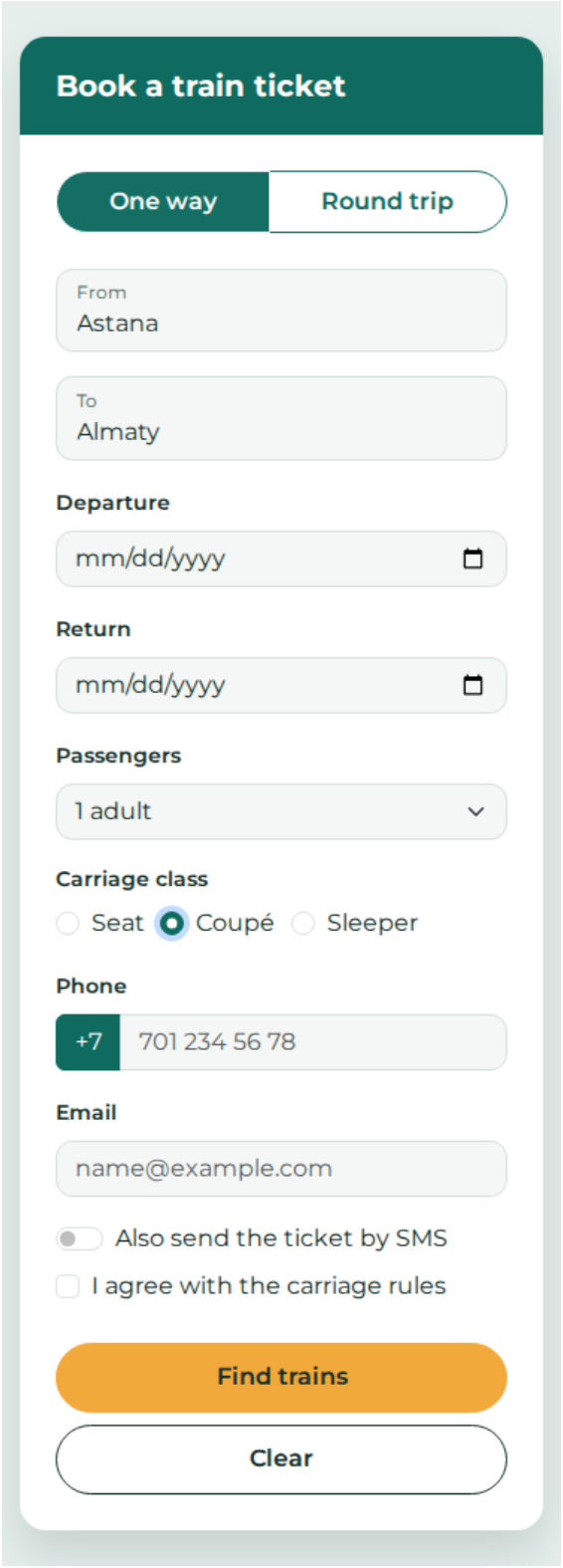


Task 1 - responsive screenshots

Laptop - viewport 1280 px



Task 1 - phone viewport 375 px



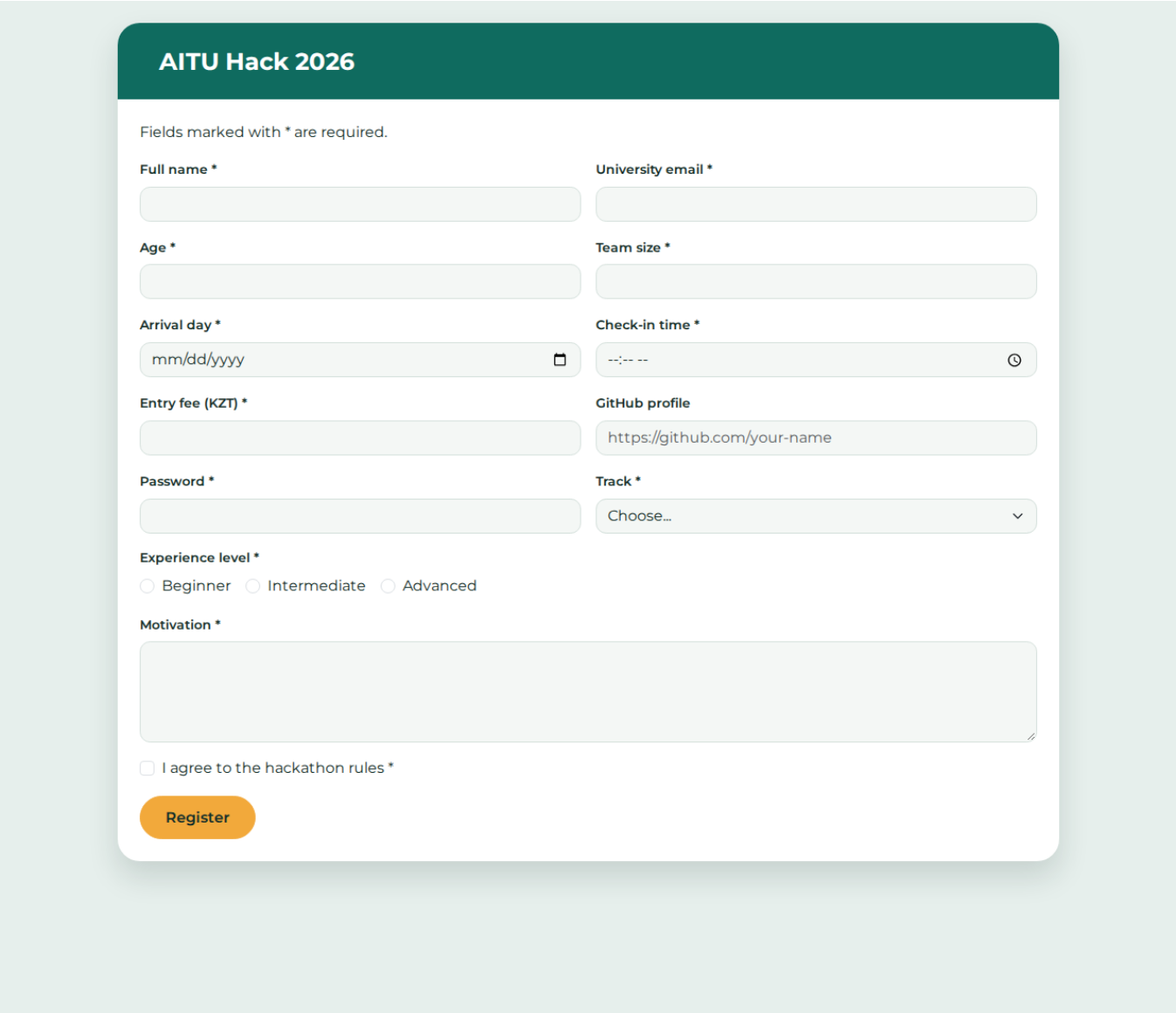
Both screenshots come from the implemented page in Chromium. At 375 px, the fields stack without horizontal overflow. The button order changes visually without moving the HTML elements. Clear was tested after entering values; text fields and radio selections reset to empty.

2. AITU Hack 2026 registration

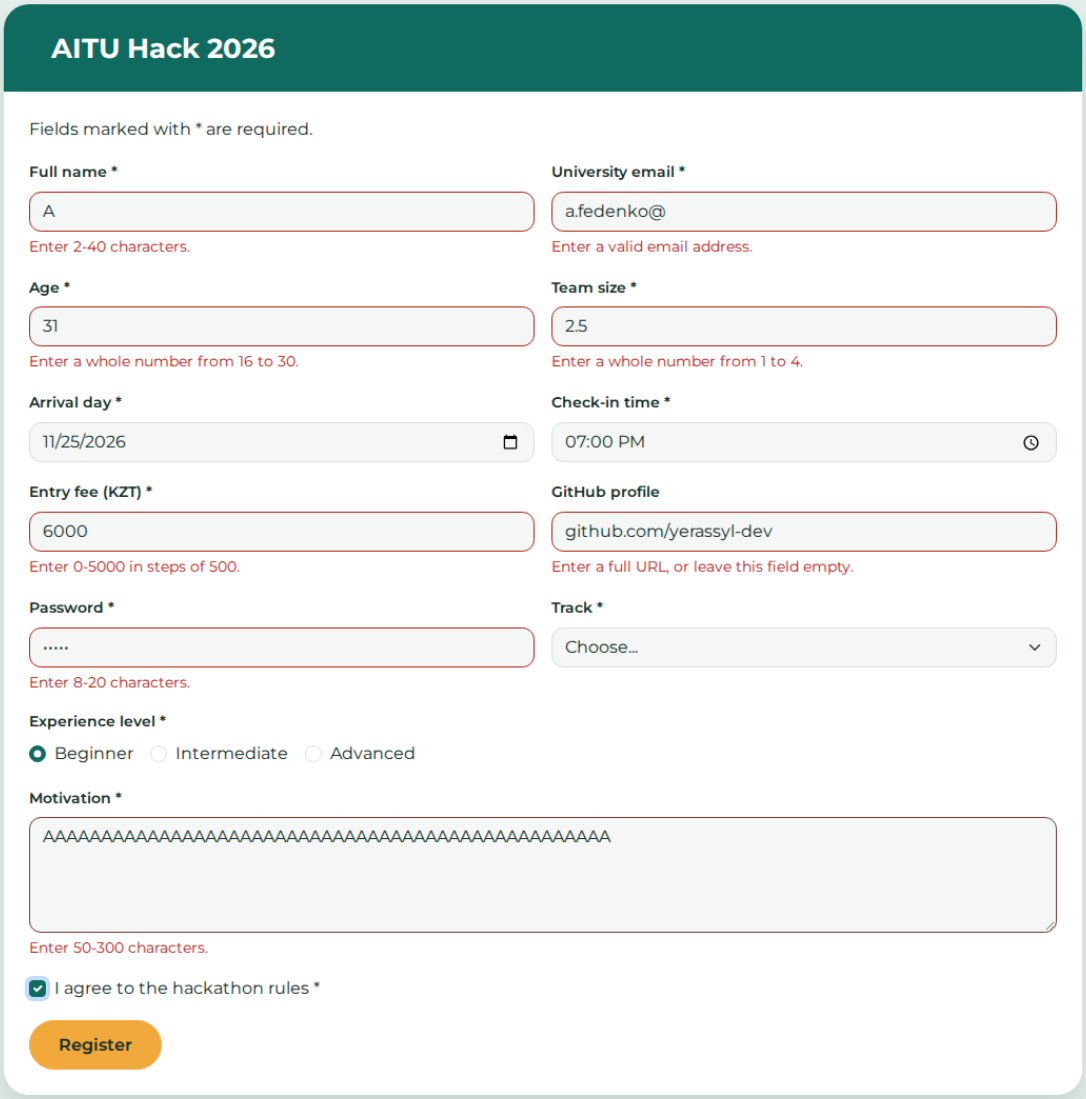
I defined validation with input types and HTML attributes only. required rejects missing values. min, max and step control numbers, dates and time. Text length is limited with minlength and maxlength. The time step is 1800 seconds, which means 30 minutes. GitHub is optional and uses type="url"; email uses type="email" without restricting the domain.

The stylesheet uses :user-invalid to show the red border and invalid-feedback after interaction or an attempted submission. On initial load the feedback is hidden. No JavaScript, pattern or novalidate is included.

Initial view - no errors



Task 2 - feedback after interaction



The following table records actual Chromium validity results. Text was entered as user input so that minlength was tested correctly. The 120-character motivation test uses 120 letter A characters; the rejection test uses 49.

These are frontend validation demonstrations. Task 2 uses POST to avoid putting a password in the URL. No backend is included, so successful validation does not create a real registration.

Task 2 - validation results

Field	Test value	Actual result
Full name	Anna-Maria Lee	Accepted
Full name	A	Rejected
University email	a.fedenko@astanait.edu.kz	Accepted
University email	a.fedenko	Rejected
University email	a.fedenko@	Rejected
Age	18	Accepted
Age	15	Rejected
Age	31	Rejected
Team size	4	Accepted
Team size	5	Rejected
Team size	2.5	Rejected
Arrival day	2026-11-21	Accepted
Arrival day	2026-11-25	Rejected
Check-in time	10:30	Accepted
Check-in time	10:15	Rejected
Check-in time	19:00	Rejected
Entry fee	1500	Accepted
Entry fee	1250	Rejected
Entry fee	6000	Rejected
GitHub profile	(empty)	Accepted
GitHub profile	https://github.com/yerassyl-dev	Accepted
GitHub profile	github.com/yerassyl-dev	Rejected
Password	Astana2026	Accepted
Password	short	Rejected
Motivation	120 characters	Accepted
Motivation	49 characters	Rejected
Track	Web	Accepted
Track	Choose...	Rejected
Experience	nothing chosen	Rejected
Experience	Beginner	Accepted
Rules agreement	not ticked	Rejected
Rules agreement	ticked	Accepted

3. Fix a bad form - problems 1-9

Line numbers below refer to the code pictured in the assignment: line 1 is the heading, line 2 is <form>, and line 21 is </form>. All original fields remain in task3.html.

Original line / code	Problem for the user	My fix
1. 3: <input type="text" placeholder="Your name">	The name field has no persistent label. Its purpose disappears after typing.	Added a visible label with for="name" and a matching id.
2. 4: <input type="text" placeholder="Email">	The browser accepts malformed email text and does not offer an email keyboard.	Used type="email", a connected label and autocomplete="email".
3. 5: <label>Phone</label>	Clicking Phone does not focus the field; the label is not associated.	Added for="phone".
4. 6: <input type="text" id="phone">	A phone number is treated as general text, making mobile entry less convenient.	Used type="tel" and autocomplete="tel".
5. 7-8: for="guests" but id="people"	The Guests label points to an element that does not exist.	Used id="guests" with for="guests".
6. 8: <input type="text" id="people">	Users can enter negative or fractional guest counts or nonnumeric text.	Used type="number", min="1" and step="1".
7. 10: type="text" placeholder="dd/mm/yy"	Date entry is ambiguous and arbitrary text is accepted.	Used type="date" with a visible Date label.
8. 10,12: id="date" occurs twice	Duplicate IDs make label targets ambiguous.	Kept date for the date field and gave the radio its own unique ID.
9. 12-13: name="inside" and name="terrace"	Both seating options can be selected at once.	Gave both radio inputs name="seating" and distinct values.

Alternative fix 1: Alternative: aria-label. I chose a visible label because all users can see it.

Alternative fix 2: Alternative: one select. I kept radios because both choices stay visible and match the original controls.

Task 3 - problems 10-17

Original line / code	Problem for the user	My fix
10. 11: Seating:	The seating choices have no semantic group name for assistive technology.	Used fieldset and legend for Seating.
11. 12-13: text after radio inputs	Inside and Terrace are not connected labels, so their text is not a clickable target.	Added a label connected to each radio ID.
12. 14: <select>	The meal selector has no accessible label.	Added a Meal label and id="meal".
13. 18: <textarea placeholder="Comments">	Comments has only disappearing placeholder text.	Added a visible connected Comments label.
14. 19: <input type="checkbox"> I agree...	The agreement text is not connected and unchecked agreement does not block submission.	Added a connected label, id, value and required.
15. 20: Reserve	A div is not a submit control and cannot be activated normally by keyboard.	Replaced it with button type="submit".
16. 3-10,14,18-19: missing name attributes	Most field values are omitted from submitted form data.	Added meaningful name attributes to every control.
17. 2-21: controls without a responsive layout	The form has no structured mobile spacing or responsive columns.	Used Bootstrap row, col-12, col-md-6, g-3 and form classes.

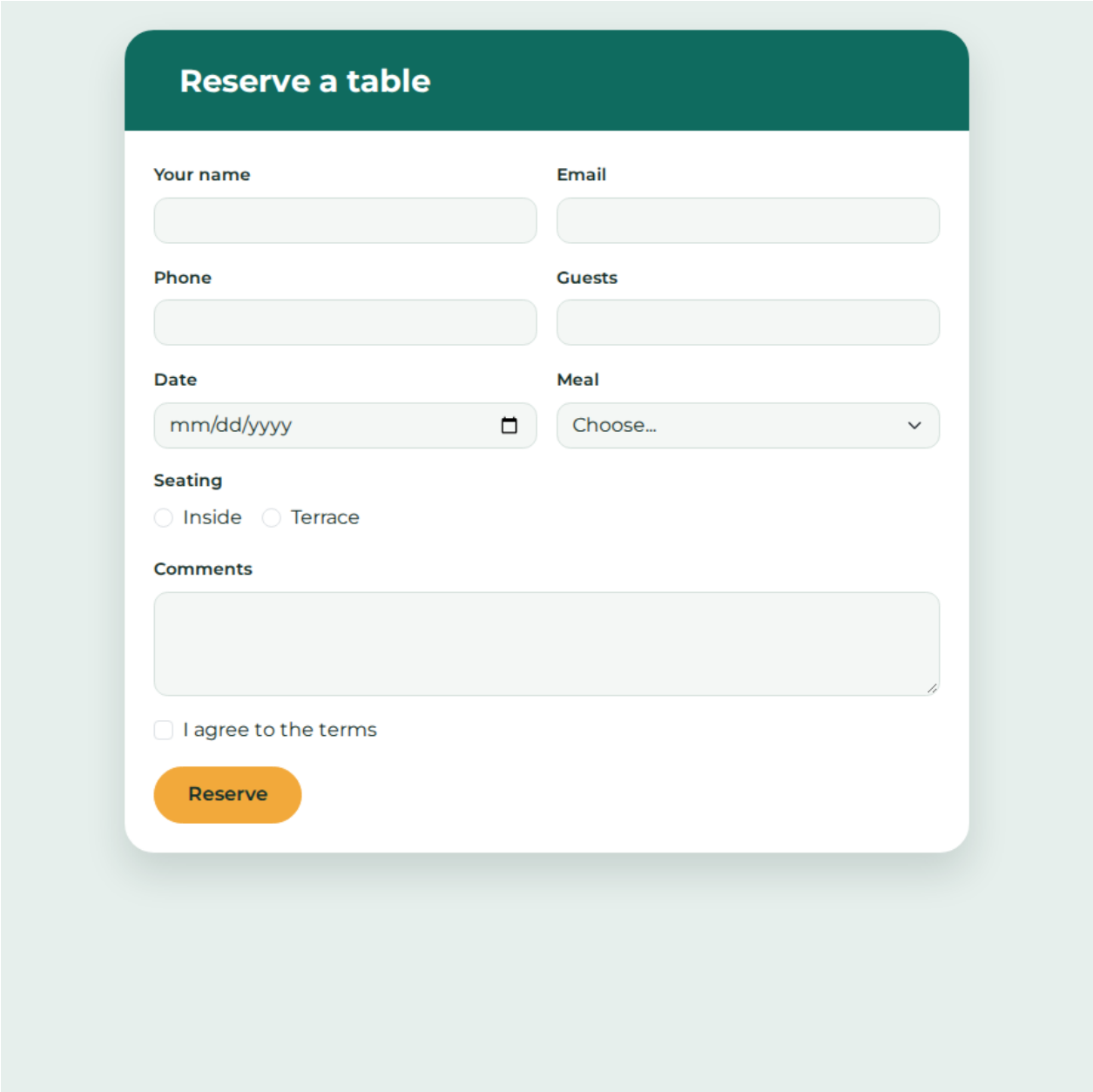
Markup validation

All three pages passed the local Nu HTML Checker with zero HTML errors. This is the checker used by the W3C HTML checking service. The hosted W3C endpoint was unavailable during the check, so the local checker was used instead.

Keyboard behavior

Native inputs, selects, radio groups, checkbox and submit button remain in normal document order. Tab reaches controls, arrow keys change radio and select choices, Space toggles checkboxes, and Enter activates the submit button. There are no positive tabindex values or clickable div replacements.

Task 3 - corrected reservation form



I used a responsive two-column layout that becomes one column on a phone. All fields have name attributes, so the browser can include their values in form data. Seating uses one radio group, and the Reserve button is a real submit button.

Conclusion

I learned how Bootstrap grid classes adapt forms to different screen sizes and how native HTML attributes validate input without scripts. I also learned that accessible labels, unique IDs, field names and semantic buttons are essential for a usable form. A good-looking control still needs correct HTML to work with the keyboard and submit data.

References

Bootstrap forms: https://getbootstrap.com/docs/5.3/forms/overview/ MDN :user-invalid: https://developer.mozilla.org/en-US/docs/Web/CSS/:user-invalid HTML checker: https://validator.w3.org/nu/